

Deep Residual Learning for Cross-Source Ground-Based Cloud Classification: Group-Aware Partitioning, Domain Harmonization, and Optimization Dynamics

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Abstract

Ground-based sky imaging provides high-spatiotemporal resolution observations of cloud fields, supporting solar irradiance forecasting, aviation meteorology, and climate modeling. However, automated classification models deployed across disparate ground sensors encounter domain shift from optical heterogeneity (regional cloud views versus wide-angle whole-sky cameras) and taxonomic discrepancies between fine-grained World Meteorological Organization (WMO) genera and operational sky-condition categories. In this paper, we evaluate these challenges across the deep residual learning family (ResNet-18, ResNet-34, and ResNet-50). First, an exact-byte duplicate audit identifies cross-split contamination in public benchmark releases (156 duplicate clusters across official GCD splits, and 3 conflicting pairs in CCSN). We apply a group-aware split protocol via Stratified Group K -Fold partitioning to prevent identical exposures from crossing partitions, defining an 80% development pool and an untouched 20% canonical test partition. Second, we define an operational five-class compatibility taxonomy mapping fine-grained WMO genera and operational sky conditions into a common label space, excluding clear sky, contrails, and mixed scenes to focus on single-cloud categories. Third, to address sample volume divergence (85.96% GCD versus 14.04% CCSN), we apply source-balanced batch sampling in expectation, supported by hyperparameter exploration on the development pool. On the canonical harmonized test partition, the joint model achieves a source-balanced accuracy of **74.58%** on ResNet-18, **74.52%** on ResNet-34, and **74.66%** on ResNet-50. Because the gap between ResNet-18 and ResNet-50 is only 0.08 percentage points, ResNet-18 serves as a practical, lightweight baseline for resource-constrained deployment. Furthermore, joint training improves classification accuracy on the sparse CCSN test component by +4.06 percentage points on ResNet-18 (56.84% to 60.90%), while cross-source evaluation reveals marked transfer asymmetry between regional and whole-sky modalities.

1 Introduction

Clouds represent the single largest source of uncertainty in global atmospheric radiative transfer, earth energy budget calculations, and numerical weather prediction models [1]. Accurate, real-time quantification of cloud types and vertical structure governs both surface solar irradiance attenuation and downward longwave greenhouse counter-radiation. While geostationary and low-Earth-orbit satellites provide synoptic continental coverage, satellite instruments suffer from coarse surface spatial resolution, sub-pixel cloud fraction ambiguity, parallax errors, and signal attenuation when observing thin boundary layer clouds underneath high cirrus decks [2].

Consequently, ground-based sky imagers have emerged as vital instruments for surface-level atmospheric observation. Ground imagers observe cloud systems directly from the surface, cap-

turing key visual features: cloud base morphology, texture, and convective geometry. Historically, cloud classification relied upon manual human synoptic observations conducted every three to six hours using World Meteorological Organization (WMO) International Cloud Atlas guidelines [3]. However, human observations remain subjective, limited in temporal frequency, and difficult to scale across remote meteorological stations.

In recent years, deep convolutional neural networks (CNNs) have demonstrated promising capability for automated ground-based cloud classification [4, 5]. Nonetheless, empirical benchmarking across heterogeneous archives encounters three practical challenges:

1. **Data Leakage via Duplicate Exposures:** Ground-based sky cameras operating on automated schedules frequently capture identical images under calm atmospheric conditions. Random splitting can distribute identical files across training and evaluation partitions, creating a risk of test-set memorization and inflated evaluation metrics.
2. **Taxonomic Divergence:** Open-access datasets follow disparate labeling schemes. The Cirrus Cumulus Stratus Nimbus (CCSN) dataset [4] adheres to WMO genera, whereas the Ground-based Cloud Dataset (GCD) [5] uses operational sky-condition classes. Direct concatenation without taxonomic harmonization leads to inconsistent supervision.
3. **Cross-Source Optical Discrepancy:** Sensor hardware varies across observation archives. CCSN images focus on localized regional cloud fields, whereas GCD captures wide-angle whole-sky views, introducing perspective, illumination, and scale domain shifts.

To evaluate these challenges, we compare ResNet-18, ResNet-34, and ResNet-50 [6] under a group-aware partitioning protocol, an operational five-class compatibility taxonomy, and source-balanced batch sampling in expectation. Supporting hyperparameter sweeps on the development pool guide model configuration, and final benchmarks are evaluated on an untouched canonical test partition with non-parametric bootstrap confidence intervals. Across sensor modalities, the joint model yields a source-balanced accuracy of 74.58% on ResNet-18 and 74.66% on ResNet-50, supporting ResNet-18 as a practical baseline for resource-constrained deployment.

2 Data Sources, Anomaly Curation, and Group-Aware Partitioning

2.1 Dataset Profiles: CCSN and GCD

To investigate cross-source domain shift, we examine two primary ground-based benchmarks:

Cirrus Cumulus Stratus Nimbus (CCSN) Dataset: Published by Zhang et al. [4], CCSN contains 2,543 raw images captured in China using digital cameras focusing on specific cloud fields. Images possess diverse resolutions and are categorized into 11 fine-grained classes: 10 WMO genera (Cirrus [Ci], Cirrocumulus [Cc], Cirrostratus [Cs], Altocumulus [Ac], Altostratus [As], Stratocumulus [Sc], Stratus [St], Cumulus [Cu], Cumulonimbus [Cb], Nimbostratus [Ns]) and one human-generated category (Contrails [Ct], 200 images). Three conflicting duplicate pairs (6 images) with identical SHA-256 file-byte hashes but conflicting genus labels were purged, leaving 2,537 verified images in the 11-class view, and 2,337 images in the 5-class harmonized benchmark after contrail exclusion.

Ground-based Cloud Dataset (GCD): Published by Liu et al. [5], GCD comprises 19,000 images collected by ground-based all-sky cameras across nine Chinese provinces during 2019–2020. Images capture wide whole-sky perspectives categorized into seven operational sky classes: 1_cumulus, 2_altocumulus, 3_cirrus, 4_clearsky, 5_stratocumulus, 6_cumulonimbus, and 7_mixed. To construct a consistent cloud-type classification benchmark, we exclude 955 mixed-sky scenes (7_mixed) and 3,739 clear-sky scenes (4_clearsky, which permits $\leq 10\%$ cloudiness per GCD authors), yielding 14,306 cloud images in the 5-class harmonized benchmark.

2.2 Observational Redundancy and Exact-Byte Duplicate Audit

In automated sky imaging systems, consecutive exposures under calm conditions frequently record byte-identical image files. An exact-byte SHA-256 file hash audit of the public GCD release identified 156 duplicate clusters spanning the official train and test directories, primarily in clear-sky and static sky conditions. Evaluating models on official splits containing cross-partition duplicates risks test-set memorization.

Furthermore, in CCSN, the audit identified three conflicting duplicate pairs (six images with identical SHA-256 file-byte hashes) assigned conflicting genus labels: two pairs conflicting between Altocumulus (*Ac*) and Altostratus (*As*), and one pair conflicting between Cirrocumulus (*Cc*) and Cirrostratus (*Cs*). These six images were removed to maintain label consistency.

2.3 Partitioning Architecture and Sample Allocation

To maintain benchmark reliability, we applied consistent data curation and partitioning rules:

1. **Purging Conflicting Duplicates:** The three conflicting duplicate pairs in CCSN were removed.
2. **Exclusion of Contrails and Mixed Scenes:** Anthropogenic contrails (*Ct* in CCSN, 200 images) and heterogeneous mixed scenes (*7_mixed* in GCD, 955 images) were excluded to focus on natural single-cloud scenes.
3. **Exclusion of Clear Sky (*4_clearsky*):** GCD contains 3,739 clear-sky images, whereas CCSN contains zero clear-sky images. Including clear sky would create an extreme domain disparity and risk shortcut learning on blue sky color; thus *4_clearsky* was excluded.
4. **Atomic Cluster Partitioning via Stratified Group K -Fold:** Exact-byte duplicate clusters share an atomic `group_id`. All partitions were generated using `StratifiedGroupKFold` (seed 42), preventing identical exposures from crossing partitions. (Note that exact-byte grouping does not identify near-duplicate scenes with minor sensor noise).

We define terminology for all data partitions as summarized in Table 1:

- **Development Pool ($N = 13,313$ images):** The 80% development pool partitioned into parameter training ($N = 10,650$) and an **internal development validation partition** ($N = 2,663$, fold 0). Validation data are used only for hyperparameter exploration and checkpoint selection; they do not constitute final benchmark evidence.
- **Canonical Test Partition ($N = 3,330$ images):** The untouched 20% evaluation partition. It decomposes into two source-specific components: the **Harmonized CCSN Test Component** ($N = 468$) and the **Harmonized GCD Test Component** ($N = 2,862$).
- **Pooled Test Partition ($N = 3,330$ images):** The sample-concatenated evaluation partition. Because GCD accounts for 85.95% of this pool ($2,862/3,330$), pooled combined accuracy is inherently GCD-heavy. Consequently, **source-balanced averaging** is adopted as the primary headline metric, treating both source domains with equal weight.

Dataset Partition / View	Classes	Train	Val (Dev)	Test	Total
CCSN 11-Class View	11	1,622	407	508	2,537
GCD 6-Class View	6	11,548	2,888	3,609	18,045
Harmonized CCSN Component	5	1,495	374	468	2,337
Harmonized GCD Component	5	9,155	2,289	2,862	14,306
Pooled Harmonized Benchmark	5	10,650	2,663	3,330	16,643

Table 1: Canonical dataset partitions and sample allocation across parameter training, internal development validation, and final test partitions.

3 Domain Harmonization: The 5-Class Compatibility Taxonomy

3.1 Operational Compatibility Formulation

The WMO International Cloud Atlas defines ten cloud genera based on physical altitude and convective structure, whereas ground-based whole-sky imagers typically aggregate fine genera into broader operational categories. This is an operational compatibility taxonomy, not a genus-preserving WMO taxonomy.

3.2 Taxonomic Mapping

Table 2 details the authoritative mapping between source labels and the five compatibility classes, canonically codified in `metadata/splits/harmonized_5bin_canonical.json`.

Harmonized Class	CCSN Genera	GCD Category	Physical Cloud Morphology
Cumulus	Cu	1_cumulus	Detached, dense convective clouds with distinct outlines and flat bases.
Alto cumulus	Ac, Cc	2_alto cumulus	Mid-level stratocumuliform rolls (<i>Ac</i>) and high-level ice ripples (<i>Cc</i>).
Cirrus	Ci, Cs	3_cirrus	High-altitude fibrous ice filaments (<i>Ci</i>) and thin veil sheets (<i>Cs</i>).
Stratocumulus	Sc, St, As	5_stratocumulus	Low continuous stratiform decks (<i>Sc, St</i>) and dense non-precipitating sheets (<i>As</i>).
Cumulonimbus	Cb, Ns	6_cumulonimbus	Deep convective towers (<i>Cb</i>) and thick precipitation-bearing layers (<i>Ns</i>).

Table 2: Taxonomic harmonization mapping CCSN and GCD into five cross-source compatibility classes.

4 Model Architecture and Orientation Invariance

4.1 Architecture Overview and Trade-offs

We evaluate three architectures from the deep residual network family [6] spanning different computational capacities:

- **ResNet-18:** 8 BasicBlocks (two 3×3 convolutions per block). Parameter count with custom head is 11,309,637 (~ 11.3 M parameters, 1.82 GFLOPs per 224×224 pass).
- **ResNet-34:** 16 BasicBlocks with deeper feature hierarchy (channel depths 64 to 512). Parameter count is 21,417,797 (~ 21.4 M parameters, 3.66 GFLOPs).
- **ResNet-50:** 16 3-layer Bottleneck blocks (1×1 , 3×3 , 1×1 convolutions) reaching 2048 channels. Parameter count is 24,034,373 (~ 24.0 M parameters, 4.12 GFLOPs).

Residual shortcut connections propagate gradients directly across layers during backpropagation, stabilizing optimization across varying depths while providing an empirical trade-off between parameter footprint and representational capacity.

4.2 Regularized Classification Head

The standard ImageNet 1,000-class fully connected layer is replaced by a regularized two-layer classification head:

$$z = \text{Linear}_{256 \rightarrow 5} \left(\text{Dropout}_{0.2} \left(\text{GELU} \left(\text{BatchNorm1d} \left(\text{Linear}_{d_{\text{in}} \rightarrow 256} \left(\text{Dropout}_{0.3}(x_{\text{pool}}) \right) \right) \right) \right) \right) \quad (1)$$

where $d_{\text{in}} = 512$ for ResNet-18/34 and $d_{\text{in}} = 2048$ for ResNet-50. Both dropout layers reside within the replacement head; no dropout is inserted into the residual backbone.

4.3 Vertical-Orientation Invariance

Ground-based cloud imagery possesses an inherent physical vertical orientation: condensation bases form horizontally at the lifting condensation level, while buoyant vertical growth extends upward. Applying vertical flipping inverts this physical structure, placing turbulent tops at the bottom and condensation bases at the top. Therefore, vertical flipping is omitted. Data augmentation is restricted to physically conservative transformations: horizontal flipping ($p = 0.5$), bounded rotation ($\pm 15^\circ$), random resized cropping (scale 0.8 to 1.0), and mild photometric jitter ($\pm 10\%$).

5 Optimization Dynamics and Hyperparameter Exploration

5.1 Differential Learning Rates and Annealing Schedule

We employ differential learning rates between the pretrained backbone and the initialized classification head: $\eta^{(\text{backbone})} = 5 \times 10^{-5}$ and $\eta^{(\text{head})} = 5 \times 10^{-4}$ (a 1:10 ratio), optimized with AdamW and decoupled weight decay $\lambda = 1 \times 10^{-2}$. Learning rates follow a cosine annealing schedule decaying to $\eta_{\min} = 10^{-6}$ over 15 epochs.

5.2 Batch Sampling and Training Disclosures

In the harmonized training split, GCD accounts for 85.96% of images ($N_{\text{GCD}} = 9,155$) and CCSN contributes 14.04% ($N_{\text{CCSN}} = 1,495$). Uniform random sampling would allow GCD samples to dominate each batch. To balance gradient updates across domains, joint training applies `WeightedRandomSampler` with inverse source-size weights:

$$w_i = \frac{0.5}{N_{\text{domain}(i)}} \quad (2)$$

This sampling procedure yields equal contributions from each source **in expectation** across iterations.

Methodological Disclosures:

1. **Training Exposure Disparity:** Joint training executes 167 batches per epoch (batch size 64) with 10,650 draws with replacement per epoch ($\sim 5,325$ CCSN draws in expectation, averaging 3.56 exposures per CCSN training image, compared to 1 draw per image across 24 batches in CCSN-only training). Observed test gains reflect the full joint recipe (expanded update budget, source-balanced draws, and cross-source features) rather than isolated feature transfer alone.
2. **Validation Loss Weighting:** While joint training samples each source with equal probability in expectation, the validation split is sample-averaged over 374 CCSN images and 2,289 GCD images ($L_{\text{val}} \approx 0.1404L_{\text{CCSN}} + 0.8596L_{\text{GCD}}$). Checkpoint selection is therefore weighted towards GCD validation loss.

5.3 Hyperparameter Exploration on Development Data

As supporting exploration, we evaluated ten hyperparameter configurations on the internal development validation partition (excluding test samples) covering AdamW and SGD with momentum, learning rates, weight decay, label smoothing, and dropout. Figure 1 displays convergence trajectories for the selected baseline configuration (Trial 08), while complete validation results across all ten screening runs are documented in Appendix Table 4.

The screening runs show that differential learning rates with AdamW ($\eta_{\text{bb}} = 5 \times 10^{-5}$, $\eta_{\text{head}} = 5 \times 10^{-4}$, Trial 08) outperform standard momentum SGD (Trial 04) by 1.2 to 2.0 percentage points across architectures (+1.5 pp for ResNet-18, +2.0 pp for ResNet-34, and +1.2 pp for ResNet-50).

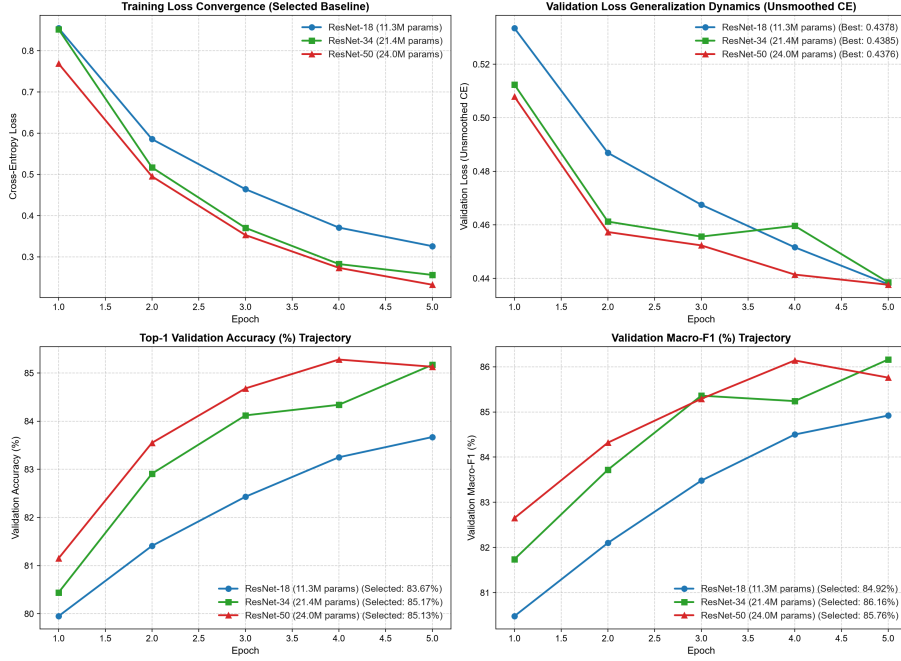


Figure 1: Gradient descent convergence dynamics across ResNet-18, ResNet-34, and ResNet-50 during 5-epoch Trial 08 screening runs on the internal development validation partition. Panels display training loss convergence (top-left), unsmoothed validation loss (top-right), validation accuracy (bottom-left), and validation macro-F1 score (bottom-right). Legends indicate checkpoint-selected screening performance.

The unregularized label smoothing setting ($\alpha = 0.0$) with weight decay $\lambda = 1 \times 10^{-2}$ yielded stable convergence across all three architectures and was selected as the baseline configuration for production benchmarks.

6 Empirical Results and Comparative Production Benchmarks

6.1 Evaluation Protocol

Production models were trained for 15 epochs on the canonical harmonized dataset using the baseline configuration. Checkpoint selection restored the state achieving minimum validation loss, which was subsequently evaluated on the canonical test partition (3,330 images). Image-level 95% bootstrap confidence intervals ($B = 1,000$) quantify sampling variability conditionally on the test partition.

6.2 Primary Empirical Headline: Source-Balanced Joint Performance

The headline benchmark result of this study is the joint CCSN+GCD model evaluated on the canonical harmonized final test partition, reported with source-balanced averaging across sensor modalities. Table 3 presents this primary comparison.

Several critical observations emerge from Table 3:

1. **Headline Performance and ResNet-18 as Practical Baseline:** ResNet-18 achieves a source-balanced accuracy of **74.58%**, compared to **74.66%** for ResNet-50—a margin of merely **0.08 percentage points**. Moreover, in balanced accuracy (macro-recall), ResNet-18 attains **73.77%**, slightly exceeding ResNet-50 (**73.44%**), and in macro-F1 ResNet-18 reaches **74.03%** versus **73.92%**. Because ResNet-18 requires 53% fewer parameters (11.3M vs. 24.0M) and 56% fewer FLOPs (1.82 vs. 4.12 GFLOPs), ResNet-18 is framed as the practical selected

Architecture	Params	CCSN ($n = 468$)	GCD ($n = 2,862$)	Source Avg (Headline)	Pooled Test ($n = 3,330$)
ResNet-18	11.3M	60.90%	88.26%	74.58%	84.41%
ResNet-34	21.4M	60.68%	88.36%	74.52%	84.47%
ResNet-50	24.0M	60.26%	89.06%	74.66%	85.02%

Table 3: Primary Benchmark Results on the Canonical Harmonized Test Partition ($N = 3,330$ images). The headline metric is the unweighted arithmetic mean of component point estimates (Source-Balanced Average), preventing GCD sample volume ($n = 2,862$) from dominating regional CCSN evaluation ($n = 468$). Bootstrap 95% confidence intervals quantify sampling variability for component and pooled holdouts in Appendix Table 5; no confidence interval is computed for the unweighted source-balanced mean, and no formal statistical equivalence is asserted for the 0.08 pp margin.

baseline for operational sky-imaging deployment, while avoiding formal claims of statistical equivalence.

2. **Primary vs. Secondary Metrics:** On the pooled test partition ($n = 3,330$), ResNet-50 attains 85.02% versus 84.41% on ResNet-18. However, because GCD contributes 85.95% of the pooled test partition (2,862 out of 3,330 images), pooled accuracy heavily reflects GCD whole-sky performance. Source-balanced averaging treats both observational modalities with equal weight, providing a more balanced evaluation of cross-source capability.
3. **Performance on Component Partitions:** On the regional CCSN test component ($n = 468$), ResNet-18 attains the highest top-1 accuracy (60.90%), while ResNet-50 achieves higher accuracy on the whole-sky GCD component (89.06%).

6.3 Cross-Source Domain Discrepancy and Transfer Asymmetry

Ground-based sky imaging archives collected under differing operational objectives exhibit substantial domain shift across camera optics, sensor perspectives, and climatological prior distributions. In the whole-sky GCD archive, Cumulonimbus represents 40.3% of images (5,764 samples), whereas in regional CCSN views, Cumulonimbus accounts for 22.1% (516 samples). Conversely, CCSN is dominated by Stratocumulus (31.2%) and features twice the relative frequency of Altocumulus (20.8% vs. 10.3% in GCD). Furthermore, GCD contains over six times as many images as CCSN (14,306 vs. 2,337), leading to substantial representation imbalance without active intervention. This empirical reality motivated the inverse-volume batch sampling weights ($w_i = 0.5/N_{\text{domain}(i)}$) defined in Section 5, providing balanced gradient signals in expectation.

This prior divergence and optical heterogeneity are consistent with the pronounced transfer asymmetry observed when single-source models are evaluated across sensor domains (Figure 2 and Appendix Table 5):

- **CCSN $\rightarrow$ GCD (Regional to Whole-Sky):** Attains 57.86% overall accuracy and 57.94% balanced accuracy on ResNet-18. Models trained on regional cloud textures retain moderate sensitivity across whole-sky scenes, achieving 75.9% recall on GCD Cumulonimbus.
- **GCD $\rightarrow$ CCSN (Whole-Sky to Regional):** Attains only 37.18% overall accuracy and 44.18% balanced accuracy. In particular, the GCD-trained model evaluated on CCSN achieves only **19.2%** Cumulonimbus recall, reflecting a substantial directional disparity compared to the 75.9% achieved in the reverse transfer direction. Models trained on whole-sky imagery are hypothesized to rely on synoptic hemispherical context and sky-dome geometry that are absent in localized telephoto frames.

While optical geometry, sensor field-of-view, and label priors are known observational

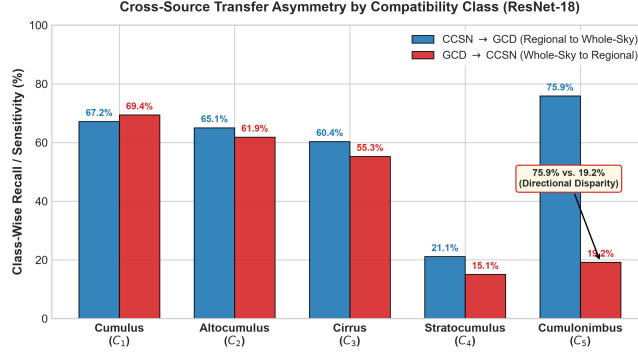


Figure 2: Cross-Source Transfer Asymmetry: Per-Class Sensitivity Disparity under Direct Cross-Source Transfer Evaluation for ResNet-18. Regional CCSN models retain moderate sensitivity across whole-sky scenes (75.9% Cumulonimbus recall on GCD), whereas whole-sky GCD models exhibit sharp directional sensitivity disparities on regional frames (19.2% Cumulonimbus recall on CCSN).

differences between the archives, attributing transfer discrepancies to specific visual features remains an observational hypothesis rather than an isolated causal claim. Detailed per-condition metrics across all 24 configurations are compiled in Appendix Table 5.

7 Error Analysis and CCSN Generalization Gain

7.1 Substantiating the +4.06 Percentage Point Gain on CCSN

As recorded in the comprehensive results (Appendix Table 5), joint training with source-balanced batch sampling improves CCSN test component accuracy from 56.84% (CCSN-only baseline) to **60.90%** on ResNet-18, delivering a gain of **+4.06 percentage points**. Gains of +2.77 pp and +1.07 pp are similarly observed for ResNet-34 and ResNet-50.

Figure 3 provides normalized confusion matrices comparing the CCSN in-domain model against the joint model evaluated on the CCSN test component (468 images).

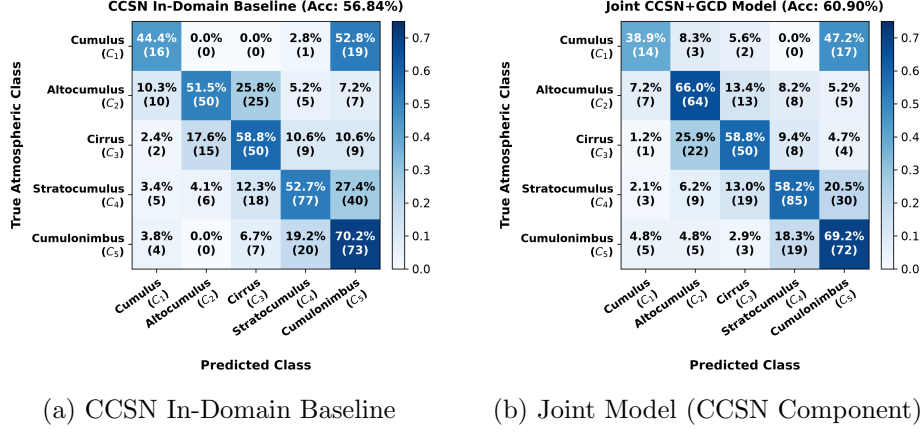


Figure 3: Normalized Confusion Matrices for ResNet-18 on the CCSN test component (468 images), showing classification distribution shifts before and after joint training with source-balanced batch sampling.

Comparison of Figures 3a and 3b shows how class-level decisions shift after joint training:

1. **Altocumulus Recognition:** Altocumulus (C_2) recall improves substantially from 51.5% to **66.0%** (+14.5 percentage points), with reduced confusion into Cirrus (from 25.8% down to 13.4%, while confusion into Stratocumulus shifts from 5.2% to 8.2%).
2. **Stratocumulus Stability:** Stratocumulus (C_4) recall increases from 52.7% to **58.2%**, consistent with improved discrimination between continuous low stratiform decks and convective rolls.
3. **High-Altitude Cirrus:** Cirrus (C_3) recall remains stable at **58.8%**, while Cumulonimbus (C_5) attains **69.2%**.

7.2 Persistent Failure Modes

Despite the improvements afforded by joint training, key meteorological ambiguities remain:

1. **Transitional Genera Ambiguity:** Altocumulus (C_2) and Stratocumulus (C_4) show persistent mutual misclassification due to shared morphological cloud roll textures during cloud layering transitions.
2. **Overcast Stratocumulus vs. Cumulonimbus:** Dense Stratocumulus decks cast dark grey diffuse lighting across ground sensors, visually resembling dark precipitation-bearing cloud bases (C_5) in single static visual frames without temporal barometric or radar data.
3. **Peripheral Horizon Clutter:** Wide-angle sky imagery contains peripheral horizon clutter (vegetation, masts, local topography) that presents non-meteorological visual features absent in regional sky views.

8 Edge Deployment Considerations

A practical motivation for selecting ResNet-18 over larger backbones is its suitability for resource-constrained meteorological stations and mobile sensing platforms. ResNet-18 requires 11.3M parameters (43.1 MiB FP32 storage) and 1.82 GFLOPs per forward pass, compared to 24.0M parameters (91.7 MiB) and 4.12 GFLOPs for ResNet-50. Because actual on-device deployment was not conducted in this benchmark study, edge latency profiling, runtime memory benchmarking, and post-training INT8 quantization across embedded architectures (such as ARM-based field stations) remain natural directions for future work.

9 Conclusion

In this work, we evaluated deep residual networks for cross-source ground-based cloud classification. By identifying cross-partition duplicate exposures in public releases and applying group-aware Stratified Group K -Fold partitioning, we set up benchmark splits that reduce the risk of exact-byte burst leakage. We defined an operational five-class compatibility taxonomy harmonizing regional digital camera and wide-angle whole-sky datasets, implemented source-balanced batch sampling in expectation, and evaluated convergence across hyperparameter configurations. Our results show that joint training achieves a source-balanced accuracy of **74.58%** on ResNet-18, within 0.08 percentage points of ResNet-50 (**74.66%**) while requiring 53% fewer parameters and 56% fewer FLOPs. Moreover, joint training improves classification accuracy on the sparse CCSN test component by +4.06 percentage points on ResNet-18, while cross-source evaluation highlights pronounced transfer asymmetry between sensor modalities. These results support ResNet-18 as an effective, computationally lightweight baseline for ground-based cloud classification.

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Appendix A: Hyperparameter Exploration on Development Data

Table 4 documents empirical validation performance across the ten hyperparameter exploration configurations evaluated on the internal development validation partition ($N = 2,663$ images). Sweeps explored differential learning rates, weight decay, label smoothing, and dropout regularization under AdamW and SGD with momentum. Trial 08 was selected as the production baseline configuration.

Trial ID	Configuration Description	Optimizer	Backbone LR	Head LR	ResNet-18	ResNet-34	ResNet-50
Trial 01	Baseline Differential LRs	AdamW	5×10^{-5}	5×10^{-4}	84.3%	83.7%	84.9%
Trial 02	Conservative (Low LR)	AdamW	2×10^{-5}	2×10^{-4}	83.3%	82.6%	83.1%
Trial 03	Aggressive (High LR)	AdamW	1×10^{-4}	1×10^{-3}	84.4%	85.0%	85.9%
Trial 04	Standard Momentum	SGD	1×10^{-3}	1×10^{-2}	82.2%	83.2%	83.9%
Trial 05	Conservative Momentum	SGD	5×10^{-4}	5×10^{-3}	81.4%	82.0%	82.4%
Trial 06	High Weight Decay (5×10^{-2})	AdamW	5×10^{-5}	5×10^{-4}	84.6%	84.9%	83.8%
Trial 07	Low Weight Decay (1×10^{-3})	AdamW	5×10^{-5}	5×10^{-4}	83.9%	83.7%	84.8%
Trial 08	Standard LRs (LS 0.0)	AdamW	5×10^{-5}	5×10^{-4}	83.7%	85.2%	85.1%
Trial 09	Heavy Dropout (0.4/0.5)	AdamW	5×10^{-5}	5×10^{-4}	85.4%	84.8%	85.7%
Trial 10	Fine Differential ($4 \times 10^{-5}/4 \times 10^{-4}$)	AdamW	4×10^{-5}	4×10^{-4}	84.5%	84.1%	85.1%

Table 4: Empirical validation accuracy across 10 hyperparameter exploration configurations on the internal development validation partition (2,663 images). Displayed values reflect 5-epoch screening runs scored on unsmoothed validation cross-entropy loss.

Appendix B: Comprehensive Production Benchmark Reference

Table 5 reports the authoritative 24-condition experimental benchmark across all architectures, single-source in-domain baselines, cross-source transfer evaluations, and joint models on the canonical harmonized test partition ($N = 3,330$ images), alongside exact 95% non-parametric percentile bootstrap confidence intervals ($B = 1,000$).

Model Architecture	Experimental Condition	Evaluation Partition	Top-1 Acc (%)	Balanced Acc (%)	Macro-F1 (%)
ResNet-18 (11.3M)	1. CCSN In-Domain Model	CCSN Test Component	56.84 [52.35, 61.32]	55.55 [50.25, 60.60]	55.01 [49.72, 60.00]
	2. Cross-Source CCSN $\rightarrow$ GCD	GCD Test Component	57.86 [56.15, 59.57]	57.94 [55.95, 59.89]	59.33 [57.48, 61.09]
	3. GCD In-Domain Model	GCD Test Component	89.34 [88.29, 90.36]	92.00 [91.02, 92.92]	91.33 [90.33, 92.22]
	4. Cross-Source GCD $\rightarrow$ CCSN	CCSN Test Component	37.18 [32.90, 41.67]	44.18 [39.48, 48.94]	36.37 [31.89, 40.63]
	5. Joint Model (CCSN+GCD)	CCSN Test Component	60.90 [56.20, 65.17]	58.23 [53.17, 63.52]	58.11 [52.78, 63.10]
	6. Joint Model (CCSN+GCD)	GCD Test Component	88.26 [87.04, 89.45]	89.32 [88.07, 90.50]	89.95 [88.74, 91.06]
	7. Joint Model (CCSN+GCD)	Pooled Test Partition	84.41 [83.21, 85.71]	84.70 [83.34, 86.03]	85.17 [83.87, 86.42]
	8. Joint Model (CCSN+GCD)	Source-Balanced Avg	74.58	73.77	74.03
ResNet-34 (21.4M)	1. CCSN In-Domain Model	CCSN Test Component	57.91 [53.42, 62.18]	52.96 [48.22, 57.86]	53.49 [48.30, 58.33]
	2. Cross-Source CCSN $\rightarrow$ GCD	GCD Test Component	30.68 [28.90, 32.42]	38.53 [36.64, 40.39]	33.74 [31.65, 35.68]
	3. GCD In-Domain Model	GCD Test Component	89.06 [87.91, 90.15]	91.35 [90.25, 92.33]	90.80 [89.69, 91.77]
	4. Cross-Source GCD $\rightarrow$ CCSN	CCSN Test Component	34.83 [30.56, 39.10]	42.37 [38.28, 46.69]	33.10 [29.03, 37.27]
	5. Joint Model (CCSN+GCD)	CCSN Test Component	60.68 [56.20, 64.96]	57.86 [53.09, 62.56]	57.64 [52.81, 62.19]
	6. Joint Model (CCSN+GCD)	GCD Test Component	88.36 [87.18, 89.55]	89.87 [88.68, 91.01]	90.07 [88.96, 91.11]
	7. Joint Model (CCSN+GCD)	Pooled Test Partition	84.47 [83.27, 85.74]	84.99 [83.73, 86.35]	85.20 [84.00, 86.50]
	8. Joint Model (CCSN+GCD)	Source-Balanced Avg	74.52	73.87	73.85
ResNet-50 (24.0M)	1. CCSN In-Domain Model	CCSN Test Component	59.19 [54.91, 63.68]	55.48 [50.77, 60.84]	55.64 [50.85, 60.67]
	2. Cross-Source CCSN $\rightarrow$ GCD	GCD Test Component	43.99 [42.17, 45.81]	52.85 [50.89, 54.82]	48.38 [46.36, 50.37]
	3. GCD In-Domain Model	GCD Test Component	89.45 [88.29, 90.57]	90.99 [89.88, 91.99]	91.22 [90.16, 92.16]
	4. Cross-Source GCD $\rightarrow$ CCSN	CCSN Test Component	33.76 [29.70, 38.25]	44.19 [40.30, 48.29]	31.97 [27.90, 36.04]
	5. Joint Model (CCSN+GCD)	CCSN Test Component	60.26 [55.77, 64.32]	56.18 [51.33, 60.90]	56.73 [51.33, 61.28]
	6. Joint Model (CCSN+GCD)	GCD Test Component	89.06 [87.84, 90.22]	90.71 [89.61, 91.82]	91.11 [90.08, 92.14]
	7. Joint Model (CCSN+GCD)	Pooled Test Partition	85.02 [83.78, 86.25]	85.46 [84.19, 86.79]	86.03 [84.86, 87.25]
	8. Joint Model (CCSN+GCD)	Source-Balanced Avg	74.66	73.44	73.92

Table 5: Comprehensive Benchmark Reference Table across all 24 Experimental Conditions on the Canonical Harmonized Test Partition (15-Epoch Production Runs with 95% Bootstrap Confidence Intervals).